

RESEARCH OVERVIEW

Quantifying Latent Policy Stance: Extracting Implicit Macroeconomic Sentiment from Central Bank Communications using Contextual Embeddings

Natural Language Processing | Macroeconomics | High-Frequency Finance

ABSTRACT

Prevailing lexicon-based and supervised sentiment methods in financial NLP are structurally ill-suited for macroeconomic policy texts, where information is conveyed not through affective polarity but through objective economic description. Central bank communications—including FOMC minutes, policy statements, and press conference transcripts—systematically employ a deliberate register of factual sterility, encoding hawkish or dovish signals in syntactic framing and contextual economic reference rather than explicit sentiment-bearing lexical tokens. This paper introduces a novel framework that abandons the positive/negative polarity paradigm in favor of mapping textual chunks to a continuous, latent hawkish-dovish policy stance dimension. We fine-tune an open-weight large language model (Llama-3 / Mistral) via Low-Rank Adaptation (LoRA) to predict high-frequency intraday movements in the 2-Year US Treasury yield as an endogenous, market-validated ground truth. Contextual embeddings extracted from the fine-tuned model's final transformer layers are used to construct a geometrically interpretable latent policy stance space. Empirical validation against basis-point shifts in yield curve instruments across FOMC event windows from 2005 to the present demonstrates significant improvement over dictionary-based baselines, offering a methodologically rigorous bridge between computational linguistics and fixed-income market microstructure.

1. A Brief History of Financial NLP: From Lexicons to Transformers

1.1 The Lexicon Era and the Loughran-McDonald Contribution

The quantitative analysis of financial text has its intellectual roots in the broader tradition of content analysis, tracing back to the manual coding schemes of communication scholars such as Berelson (1952) and the subsequent computational operationalization of those methods. Within finance, the decisive methodological inflection point arrived with Tetlock (2007), whose seminal *Journal of Finance* paper demonstrated that the fraction of negative words in *Wall Street Journal* columns—operationalized via the Harvard IV-4 General Inquirer dictionary—predicted aggregate market returns with statistically significant power. This work established the canonical paradigm: treat financial documents as bags-of-words, count sentiment-bearing tokens against a pre-specified lexicon, and regress the resulting scalar score against an asset price outcome.

The central limitation of applying general-purpose lexicons to financial text was identified decisively by Loughran and McDonald (2011) in their landmark *Journal of Finance* study. They demonstrated that words classified as negative in the Harvard IV-4 lexicon—such as "tax," "liability," and "cost"—appear with high frequency in financial disclosures without carrying the negative connotations implied by the general-language dictionary. Loughran and McDonald (LM) constructed a purpose-built financial sentiment word list by training their classification on 10-K annual filings, dramatically improving the predictive power of word-list methods for corporate disclosure analysis. Their six-category lexicon (Negative, Positive, Uncertainty, Litigious, Strong Modal, and Weak Modal) became the de facto standard for the analysis of SEC filings, earnings call transcripts, and related corporate communications, and continues to be widely deployed in empirical accounting and finance research.

"Words that humans would consider to be negative (e.g., 'tax,' 'liability,' 'cost') often have a neutral connotation in the context of finance." — Loughran & McDonald (2011), Journal of Finance.

The LM paradigm proved highly productive for corporate finance applications. Subsequent research by Huang, Teoh, and Zhang (2014) on earnings conference call tone, García (2013) on newspaper pessimism and stock returns, and Buehlmaier and Whited (2018) on financial constraints language extended the lexicon methodology across a wide range of empirical settings. Machine learning refinements—including term-frequency/inverse-document-frequency weighting and topic modeling via Latent Dirichlet Allocation—added dimensionality to the bag-of-words representation without fundamentally departing from the discrete-token counting paradigm.

1.2 The Neural Turn: BERT, FinBERT, and Contextualized Representations

The introduction of the Transformer architecture by Vaswani et al. (2017) and the subsequent development of pre-trained bidirectional language models, most notably BERT (Devlin et al., 2019), catalyzed a fundamental paradigm shift in NLP. Unlike static word embeddings such as Word2Vec (Mikolov et al., 2013) or GloVe (Pennington et al., 2014), BERT generates contextually conditioned token representations through multi-head self-attention operating over the full input

sequence. The practical consequence for sentiment analysis is significant: the same lexical token receives a different embedding—and hence a different inferred sentiment contribution—depending on its sentential context, a property that static lexicons are architecturally incapable of replicating.

The finance-specific pre-trained language model FinBERT (Yang, UY, and Huang, 2020; Araci, 2019) adapted the BERT architecture to the financial domain through continued pre-training on a large corpus of financial news articles and subsequent fine-tuning on a manually annotated sentiment classification task. FinBERT substantially outperformed LM lexicon baselines on financial news sentiment classification benchmarks, demonstrating the value of domain-adaptive pre-training for capturing the distributional semantics of financial vocabulary. Extensions such as FinBERT-Tone further refined the model for nuanced polarity detection in earnings calls.

Parallel developments in economic NLP include the application of topic models and supervised classifiers to central bank communications. Picault and Renault (2017) applied an LDA-based approach to ECB communications to estimate topic-specific hawkishness; Ehrmann and Talmi (2020) used text similarity metrics to measure deviation from prior ECB statements as a proxy for communication surprise. Hansen, McMahon, and Prat (2018) applied computational text analysis to Federal Reserve FOMC transcripts to measure deliberation effects. These contributions notwithstanding, none of the prevailing methodologies in the macroeconomic NLP literature has deployed the full representational capacity of modern large language models with market-validated, continuous-valued supervision—the specific methodological gap this paper is designed to address.

1.3 Structural Limitations of Existing Methods in Macroeconomic Contexts

The inadequacy of lexicon-based and standard supervised sentiment methods for macroeconomic policy text is not merely an empirical matter of performance degradation—it reflects a fundamental theoretical mismatch between the representational assumptions of those methods and the communicative structure of central bank language. Lexicon methods require sentiment to be carried by discrete tokens; supervised sentiment classifiers, including FinBERT, are trained to identify explicit positive or negative polarity. Central bank communications are, by design and institutional convention, written to be semantically neutral at the surface level. The informational content of these documents is encoded not in sentiment-bearing lexical items but in the selection and framing of economic facts, the choice of verb aspect and modal auxiliaries, the relative emphasis accorded to inflationary versus real-activity considerations, and the deployment of specific technical economic terminology within a shifting macroeconomic context.

The Sterile Language Problem

A factual statement such as "the labor market remains historically tight" contains no tokens classified as positive or negative by any standard financial lexicon. Yet in the context of above-target inflation and a rate-tightening cycle, this clause unambiguously signals a hawkish policy disposition—that the Federal Reserve views labor market slack as insufficient to warrant rate cuts. Standard NLP methods, operating on surface-level polarity, are blind to this inferential layer. This represents the central theoretical deficiency this research addresses.

2. The Research Gap: Implicit Policy Stance and the Sterile Language Problem

2.1 Explicit Polarity versus Implicit Macroeconomic Stance

The theoretical gap motivating this research can be precisely articulated as the distinction between explicit semantic polarity and implicit macroeconomic policy stance. Explicit polarity refers to the affective valence that can be inferred from the surface-level lexical content of a text without reference to domain-specific macroeconomic context. Implicit policy stance, by contrast, refers to the latent disposition of a central bank—toward tighter or looser monetary conditions, higher or lower interest rates—that is encoded in the factual and descriptive content of its communications as interpreted by agents with a model of the macroeconomy.

This distinction maps onto the broader NLP distinction between sentiment analysis and stance detection, the latter of which has received growing theoretical attention (Küçük and Can, 2020). However, the specialized sub-problem of macroeconomic policy stance detection is distinguished by two additional complicating features. First, the relevant interpretive frame is not a social or political debate but a complex dynamic macroeconomic environment—specifically, the state of inflation, the output gap, and the expected path of the policy rate—relative to which any given factual description acquires its hawkish or dovish character. Second, the institutional communicators (the Federal Reserve, ECB, Bank of Japan) have explicit incentives and established practices of communicative restraint: they signal deliberately through understatement, through shifts in word choice that are imperceptible to naive linguistic analysis but consequential to trained market participants.

2.2 Technical Inadequacy of Dictionary and Early Transformer Methods

From a technical standpoint, four specific failure modes characterize existing methods when applied to macroeconomic policy texts:

- Context-blindness of lexicon methods: The LM lexicon assigns fixed polarities to tokens irrespective of sentential context, macroeconomic regime, or document-level discourse structure. It cannot represent the fact that identical phrasing about labor market tightness carries diametrically opposite policy implications depending on whether the economy is in a disinflationary or inflationary regime.
- Binary discretization: Both lexicon-based scores and FinBERT-style classifiers map text to a small number of discrete categories (positive/negative/neutral or hawk/dove). The policy stance of central bank communications is, in practice, a continuous latent quantity—a given statement may be "moderately hawkish relative to market expectations" or "marginally more dovish than the previous meeting"—and this continuous structure is destroyed by categorical discretization.
- Annotator-dependent supervision: Datasets such as the HCB (Hawkish/Centrist/Bearish) corpus annotated by human coders introduce annotator disagreement, temporal inconsistency, and the inability to capture the interaction between text content and contemporaneous macroeconomic conditions. Human annotators cannot integrate the full state of the macro environment at the time of each statement, whereas financial market prices do so automatically.
- Out-of-distribution generalization failure: Models fine-tuned on corporate earnings call sentiment do not transfer to FOMC communications because the domains differ not only in vocabulary distribution but in the fundamental communicative pragmatics—the way linguistic form maps onto institutional intent.

2.3 The Yield Curve as a Macroeconomic Reading of Text

A key theoretical contribution of this paper is the reframing of the ground-truth supervision problem. Rather than asking what a human annotator interprets a passage to mean, we ask: what does the US Treasury market—populated by professional fixed-income investors with comprehensive macroeconomic models—"read" this passage to imply for the expected path of the federal funds rate? The 2-Year US Treasury yield is the financial market's continuous, real-time aggregate forecast of Fed policy over the near-term horizon. Its intraday movement in a narrow window around FOMC communication releases constitutes an objective, incentive-compatible measurement of the latent policy stance encoded in those communications. This endogenous market validation approach represents a methodologically superior alternative to any form of human annotation or lexicon-based approximation.

3. Methodological Framework: A High-Frequency Endogenous Embedding Pipeline

3.1 Data Construction and Preprocessing

The corpus for this study spans the complete public record of FOMC communications from January 2005 through the present, comprising three primary document types: (1) post-meeting policy statements, typically 300–600 words in length, released immediately following FOMC decisions; (2) meeting minutes, released with a three-week lag, typically 4,000–8,000 words; and (3) post-meeting press conference transcripts (available from 2011 onward following the FOMC's adoption of quarterly press conferences), typically 10,000–20,000 words. This corpus encompasses approximately 240 discrete FOMC event windows, spanning multiple monetary policy regimes including the post-GFC zero lower bound period (2009–2015), the subsequent gradual normalization cycle (2015–2018), the COVID-19 emergency accommodation (2020–2021), and the aggressive tightening cycle (2022–2024).

Documents are segmented into paragraph-level and sentence-level chunks using a constituency-aware segmentation algorithm that preserves topical coherence within chunks. Each chunk is annotated with its source document type, FOMC meeting date, and a unique chunk identifier. For press conference transcripts, speaker attribution is preserved to allow differentiation between prepared remarks (Chair's opening statement) and extemporaneous responses to journalist questions—a distinction with potential policy signal value.

The high-frequency market label construction proceeds as follows. For each FOMC event, we obtain tick-level data for the 2-Year US Treasury yield (via CME Group 2-Year Treasury Note futures) and S&P 500 E-mini futures from a licensed high-frequency data provider. We define a narrow event window of $[t-10 \text{ minutes}, t+30 \text{ minutes}]$ relative to the documented release timestamp of each FOMC communication. The primary dependent variable is Δy_2 , the basis-point change in the 2-Year yield from the pre-release reference price to the peak post-release price within the event window, computed as a signed quantity (positive for yield increases, i.e., hawkish surprises; negative for yield decreases, i.e., dovish surprises). This label construction approach follows the identification strategy of event study designs in the monetary policy transmission literature (Gürkaynak, Sack, and Swanson, 2005; Nakamura and Steinsson, 2018).

Why High-Frequency Market Data Outperforms Human Annotation as Ground Truth

Human annotators face three fundamental limitations as supervisory signals for this task: (1) they cannot integrate the full contemporaneous macroeconomic state vector into their interpretation of a passage; (2) they apply inconsistent interpretive frames across the multi-decade temporal span of

the corpus; and (3) their annotations are ordinal categorical rather than continuous. The 2-Year Treasury yield aggregates the interpretations of thousands of professional fixed-income investors, each with access to the full information set—macroeconomic forecasts, prior FOMC communications, real-time economic data—and with direct financial incentives for accuracy. It is, in the language of mechanism design, an incentive-compatible, continuously-valued, exogenous revelation of the market's latent hawkish/dovish reading of each document.

3.2 Model Architecture: LoRA Fine-Tuning of Open-Weight LLMs

The choice of base model architecture reflects a deliberate prioritization of zero-shot economic reasoning capacity over domain-specific pre-training. We select from among the current generation of open-weight instruction-tuned LLMs—specifically Llama-3-8B-Instruct (Meta AI, 2024) and Mistral-7B-Instruct-v0.3 (Mistral AI, 2024)—on the grounds that their emergent capabilities for contextual inference, economic reasoning, and long-range dependency resolution substantially exceed those of BERT-scale encoder models, including FinBERT and its derivatives. The 7B–8B parameter scale represents the empirically established sweet spot for instruction-following and reasoning capacity within the class of models that can be efficiently fine-tuned on a single high-memory GPU (e.g., NVIDIA A100 80GB).

Full fine-tuning of a 7B–8B parameter model on the available corpus size (~5,000–15,000 labeled text chunks after segmentation) would be inadvisable on both computational and statistical grounds: the number of trainable parameters would vastly exceed the labeled data points, inducing catastrophic overfitting. We therefore employ Low-Rank Adaptation (LoRA; Hu et al., 2022), a parameter-efficient fine-tuning method that freezes the pre-trained model weights and introduces trainable low-rank decomposition matrices into the attention projection layers. Specifically, for weight matrix $W_0 \in \mathbb{R}^{(d \times k)}$, LoRA parameterizes the update as $\Delta W = BA$, where $B \in \mathbb{R}^{(d \times r)}$ and $A \in \mathbb{R}^{(r \times k)}$ with rank $r \ll \min(d, k)$. The effective number of trainable parameters is reduced by a factor of $d \cdot k / (r \cdot (d + k))$, typically by 100× or more relative to full fine-tuning, while empirically preserving near-full-fine-tuning performance across diverse tasks (Hu et al., 2022).

The fine-tuning objective is a regression task: given a text chunk x , the model predicts the continuous basis-point yield change Δy_2 associated with the containing FOMC event. A linear regression head is appended to the final transformer layer's [CLS]-equivalent token representation, and the composite model is trained end-to-end (frozen base weights + LoRA adapter matrices + regression head) with mean-squared error loss. This formulation is critical: by training the model to predict a continuous market reaction rather than a discrete sentiment category, we preserve the full informational resolution of the yield signal and avoid the information loss inherent in categorization.

3.3 Contextual Embedding Extraction and Latent Stance Space

Following fine-tuning, the regression head is detached, and the model is deployed as a fixed-weight feature extractor. For each text chunk x , we extract the contextual embedding $e(x) \in \mathbb{R}^d$ from the final transformer layer, where d is the model's hidden dimension (4,096 for Llama-3-8B; 4,096 for Mistral-7B). These embeddings encode the model's full contextual representation of the chunk, informed by the fine-tuning objective that has re-calibrated the internal representations to be predictive of market-validated policy stance.

The resulting embedding matrix $E \in \mathbb{R}^{(N \times d)}$ —where N is the total number of text chunks—is subjected to dimensionality reduction via principal component analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP; McInnes et al., 2018) to construct a geometrically interpretable two- or three-dimensional latent policy stance space. We hypothesize that the first principal component of this embedding space will align with the hawkish-dovish policy dimension, as validated by its correlation with the Δy_2 labels. Secondary components may capture orthogonal dimensions of variation such as economic uncertainty acknowledgment, forward guidance specificity, or financial stability concern.

The latent stance score for any given chunk is then defined as the projection of its embedding onto the first principal component vector: $s(x) = e(x)^T \cdot v_1$, yielding a scalar continuous index of implicit policy hawkishness. This score is interpretable, comparable across documents and across time, and directly validated against market reactions. The paper demonstrates the utility of this score in multiple empirical applications: predicting yield curve movements out-of-sample, characterizing communication regime shifts at turning points in the policy cycle, and decomposing document-level stance as the aggregated stance of constituent paragraph chunks.

3.4 Validation and Benchmark Comparisons

Methodological rigor demands systematic comparison against competitive baselines. The proposed framework is benchmarked against: (i) LM negative word frequency; (ii) LM net sentiment (positive minus negative); (iii) the Hawk-Dove-Neutral (HDN) classifier of Apel and Grimaldi (2012); (iv) FinBERT zero-shot inference; (v) a vanilla BERT model fine-tuned on the same LoRA training objective; and (vi) an autoregressive GPT-3.5 API-based zero-shot chain-of-thought hawkishness scorer. Evaluation metrics include: Pearson and Spearman correlation of the scalar stance score with Δy_2 ; directional accuracy (sign prediction); mean absolute error in basis points; and F1 score on a held-out set of binary hawk/dove annotations drawn from existing labeled datasets.

4. Future Research Directions

4.1 Multimodal Policy Stance: Prosodic and Paralinguistic Analysis of Press Conferences

The methodological framework developed in this paper treats central bank communications as purely textual artifacts. Yet FOMC press conferences, ECB press conferences, and analogous events at other major central banks are audio-visual communications in which substantial information may be encoded in the paralinguistic and prosodic features of the speaker's delivery—vocal pitch contour, speech rate variation, pause duration, vocal tremor, and affective tone—rather than, or in addition to, the lexical content of the transcript. The emerging subfield of multimodal financial analysis provides both the motivation and the technical scaffolding for extending the present framework to this dimension.

A multimodal extension would involve: (1) extraction of raw audio features from archived press conference recordings using automatic speech processing tools (Wav2Vec 2.0, HuBERT); (2) estimation of continuous prosodic features including fundamental frequency (F0), intensity, speaking rate, and jitter/shimmer; (3) fine-tuning a multimodal transformer architecture that fuses textual and acoustic representations to jointly predict Δy_2 . The central hypothesis, motivated by research in affective computing (Schuller et al., 2013) and political communication (Gidron and Ziblatt, 2019), is that prosodic features carry orthogonal information to textual content—that a Fed Chair's vocal hesitancy when answering questions about inflation expectations, for example, encodes uncertainty not captured in the words spoken. If confirmed, this extension would not only improve predictive performance but would open a research program on the psychophysiology of central bank communication that has no precedent in the monetary economics literature.

4.2 Cross-Central Bank Generalization and Linguistic Transfer

The United States Federal Reserve is one of more than 150 central banks globally, each operating in a distinct institutional, linguistic, and macroeconomic context. A productive extension of this framework would examine the extent to which the latent policy stance representations learned from FOMC communications generalize—or require domain-specific adaptation—when applied to the communications of the European Central Bank (ECB), the Bank of Japan (BOJ), the Bank of England (BOE), and the People's Bank of China (PBOC).

This research direction raises theoretically significant questions about the cross-lingual and cross-institutional transferability of macroeconomic policy semantics. For central banks that communicate primarily in languages other than English (ECB in German and French as well as English, BOJ in Japanese, PBOC in Mandarin), the analysis requires multilingual language models such as mBERT or XLM-RoBERTa, or language-specific adaptations. The corresponding high-frequency market labels would be drawn from Euro-area sovereign yield spreads (for ECB), Japanese Government Bond yields (for BOJ), and relevant domestic yield instruments.

Comparative analysis across central banks could yield insights into the degree to which hawkish/dovish linguistic conventions are institutionally universal versus idiosyncratic—a question with implications both for international policy coordination and for the theory of central bank communication.

4.3 Real-Time Deployment and Automated Fixed-Income Trading Algorithms

The latent policy stance score derived from this framework is, in principle, computable in near-real-time during the release of FOMC documents—a property that creates a direct research bridge to the literature on algorithmic news trading and textual alpha generation. A natural extension involves integrating the inference pipeline into a real-time event-driven trading system operating on fixed-income instruments. The system architecture would involve: (1) a document ingestion layer that monitors the FOMC's official release channels with sub-second latency; (2) a GPU inference server that computes paragraph-level stance embeddings and aggregates them to a document-level hawkishness score within the low-latency processing constraints; (3) a signal translation module that maps the continuous stance score to a target duration position in 2-Year Treasury futures; and (4) an order execution layer interfacing with electronic trading venues.

Critical empirical questions for this research direction include: what is the achievable inference latency of the fine-tuned model, and does it allow systematic participation in the initial yield repricing, or only in the subsequent sustained repricing? How does the information in the model's stance score decay over the minutes and hours following a release—i.e., what is the effective shelf life of the textual alpha? Is the relationship between stance score and yield movement structurally stable across monetary policy regimes, or does the mapping require online recalibration? These questions connect the NLP methodology to the market microstructure literature on price discovery and the speed of information incorporation (O'Hara, 2003).

4.4 Temporal Dynamics of Policy Language and Regime Change Detection

A further high-value research direction concerns the temporal evolution of the latent policy stance space itself. The present framework treats the mapping from text to stance as stationary across the full sample period (2005–present), but this assumption is almost certainly violated across the major monetary policy regimes in the sample—the pre-GFC tightening cycle, the zero-lower-bound accommodation period, the taper tantrum episode, the normalization cycle, the COVID emergency, and the post-COVID inflation shock. Each of these regimes is characterized by distinct communicative conventions, evolving technical vocabulary (e.g., the introduction of "forward guidance" as a distinct policy tool, the emergence of "quantitative tightening" as a concept), and potentially different mappings from factual economic description to market-implied policy stance.

A rigorous investigation of this temporal structure could employ: (i) rolling-window re-estimation of the LoRA adapter to track how the latent stance space evolves month by month; (ii) Bayesian structural break detection on the time series of aggregate stance scores to identify regime changes endogenously; (iii) visualization of the trajectory of the FOMC's centroid in the latent policy stance space over the full sample, providing a geometrically interpretable history of the Federal Reserve's evolving communication strategy. This line of inquiry would constitute a novel contribution to the macroeconomic history literature—specifically to the study of how central banks learn to communicate, how their language adapts under pressure from external shocks, and how markets update their interpretive models of central bank behavior.

Key References and Foundational Literature

The following works constitute the core theoretical and methodological literature underpinning this research framework:

Author(s)	Venue	Title
Loughran & McDonald (2011)	Journal of Finance	When is a Liability Not a Liability? Textual Analysis, Dictionaries, and 10-Ks
Tetlock (2007)	Journal of Finance	Giving Content to Investor Sentiment: The Role of Media in the Stock Market
Devlin et al. (2019)	NAACL-HLT	BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding
Yang, UY & Huang (2020)	arXiv	FinBERT: A Pretrained Language Model for Financial Communications
Hu et al. (2022)	ICLR	LoRA: Low-Rank Adaptation of Large Language Models
Vaswani et al. (2017)	NeurIPS	Attention Is All You Need
Gürkaynak, Sack & Swanson (2005)	American Economic Review	The Sensitivity of Long-Term Interest Rates to Economic News
Nakamura & Steinsson (2018)	Quarterly Journal of Economics	High-Frequency Identification of Monetary Non-Neutrality
Hansen, McMahon & Prat (2018)	American Economic Review	Transparency and Deliberation within the FOMC
Picault & Renault (2017)	Journal of International Money and Finance	Words are Not All Created Equal: A New Measure of ECB Communication
McInnes et al. (2018)	arXiv	UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction
Küçük & Can (2020)	IEEE Access	Stance Detection: A Survey

*This document constitutes a pre-submission research overview for internal review and peer consultation.
All methodological claims are subject to empirical validation in the final manuscript.*